Similar pottery, step by step: one bowl photo

Version 2026-11-07 · the 🏺 Similar pottery page of Glaze Search, driven in a browser through five scenarios with one photo, from simple to complicated. Every screenshot is the real page; every number is the page's own result (a test copy of the app with its own logs).

▭ red frame = where to look in the screenshot (the inside stays see-through) · purple = done by the text branch (`vendor/gc2_text_branch`) · the colour-match model of `test_similarity.zip` is **not** used on this page: the pot search uses the project's first recipe model and the photo models.



The photo used in scenarios 2–5: a glossy dark-blue bowl seen from above, lighter blue centre, white window reflections. It is not a Glazy photo, so its own recipe is not in the database: the results are pots that look alike.

What the page does

It compares your object with **12,309 objects** cut out of Glazy photos and lists the most similar pots. Each pot gets one similarity per **component**:

Component	Compares	How
glaze appearance	the look of the glaze	DINOv2 (frozen) on the cut-out → the first recipe model's trained image head → cosine
shape	the outline only	0.5 × outline measures + 0.5 × DINOv2 on the black silhouette (no colour)
surface / effects	finish, transparency, effects, defects	the attribute model's probabilities, Hellinger affinity
colour regions	every colour area and where it is	CIEDE2000 between matched regions + a position penalty
text (optional)	your words	cleaned into the training format → the first recipe model's text embedding → cosine
formula (optional)	your oxides	the first recipe model's formula embedding → cosine

How a pot's score is made

$z = (\text{similarity} - \text{average over all 12,309 pots}) \div \text{spread}$ · $\text{score} = \Sigma\ \text{share} \times z$

z says how far above an ordinary pot this one is, for that component and this search. The **shares**: the photo part gets 0.40, split by the preset (Balanced: appearance 0.30 · shape 0.20 · surface 0.15 · colour 0.35); words add 0.35 and a formula 0.25; all are divided by their total so they add up to 1. So the score is **not 0–1**: **0 is an average pot**, 2 means about two spreads above average. The card's *Why this rank?* shows the similarity per component and the share × z (the "contribution"); the average and spread are not on the page: they are recovered below from the page's own numbers.

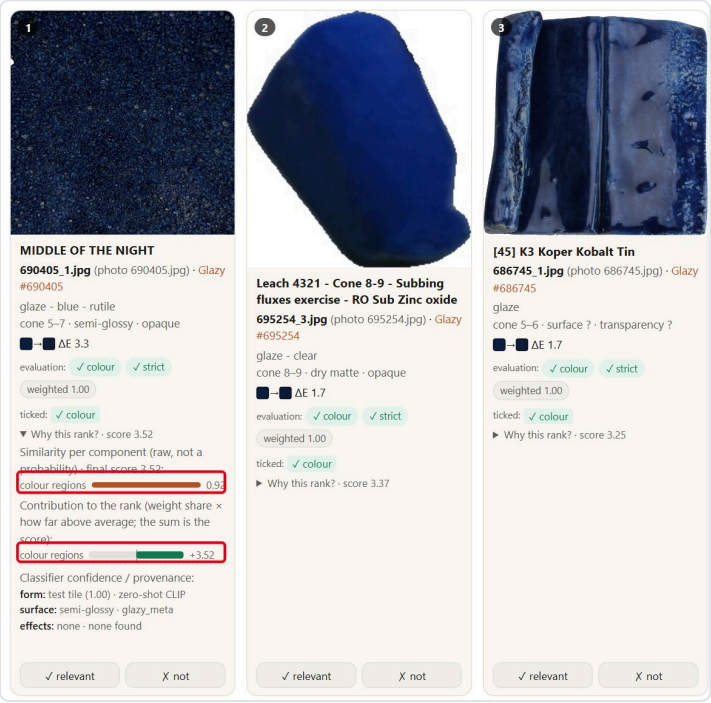
Colour regions in detail: for a query region i and a pot region j, cost = min(1, ΔE00 / 40) + 0.5 × (distance between the region centres). Each query region takes its cheapest pot region and each pot region its cheapest query region, weighted by importance (distinctiveness + area + contrast); similarity = 1 − (the two averages) / 2.

Scenario 1 · Only a colour (no photo)

1. Open 🏺 Similar pottery with no photo.
2. Card 4 · Colours: type the HEX `#0E1F3E` (navy) and press **Add**; "typed / picked" is selected.
3. Press **Search**.



Card 4: the typed colour (framed) is the only input.



The first three pots; #1 opened at *Why this rank?*: one component only.

What was computed

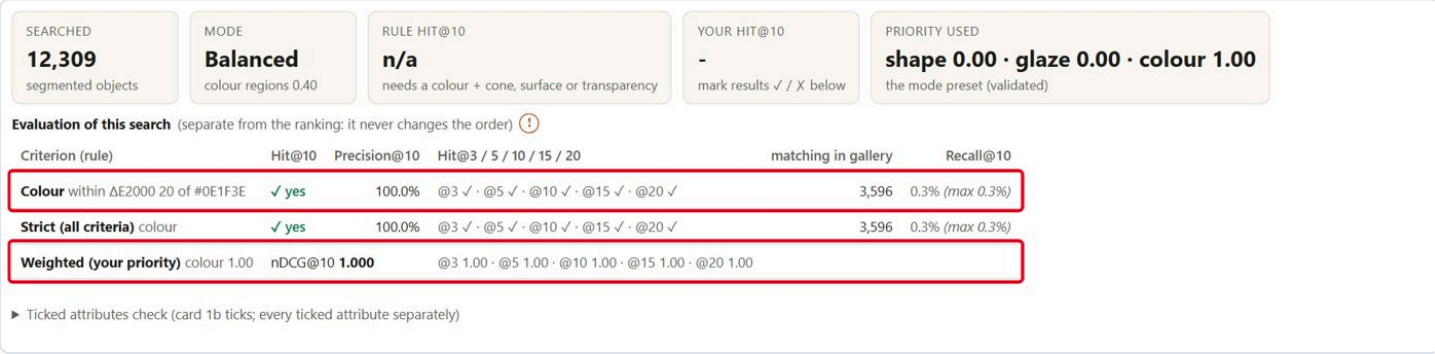
Only **colour regions** exists, so its share is 1.00 and the score is simply its z.

component	similarity	average	spread	z	share	share × z
colour regions	0.919	0.227	0.197	3.52	1.000	**+3.517**
score of #1 MIDDLE OF THE NIGHT						**3.517**

- **MIDDLE OF THE NIGHT** · 3.52
- **Leach 4321 - Cone 8-9 - Subbing fluxes exercise - RO Sub Zinc oxide** · 3.37
- **[45] K3 Koper Kobalt Tin** · 3.25

nDCG@10 **1.00** · Precision@10: shape – · glaze – · colour 100 % · strict 100 % · with no photo only the colour rule can be checked, so "strict" = colour.

What it shows: every result has your colour, but nothing asked for a shape or a surface: a tile, a test strip and a bowl can all come first.



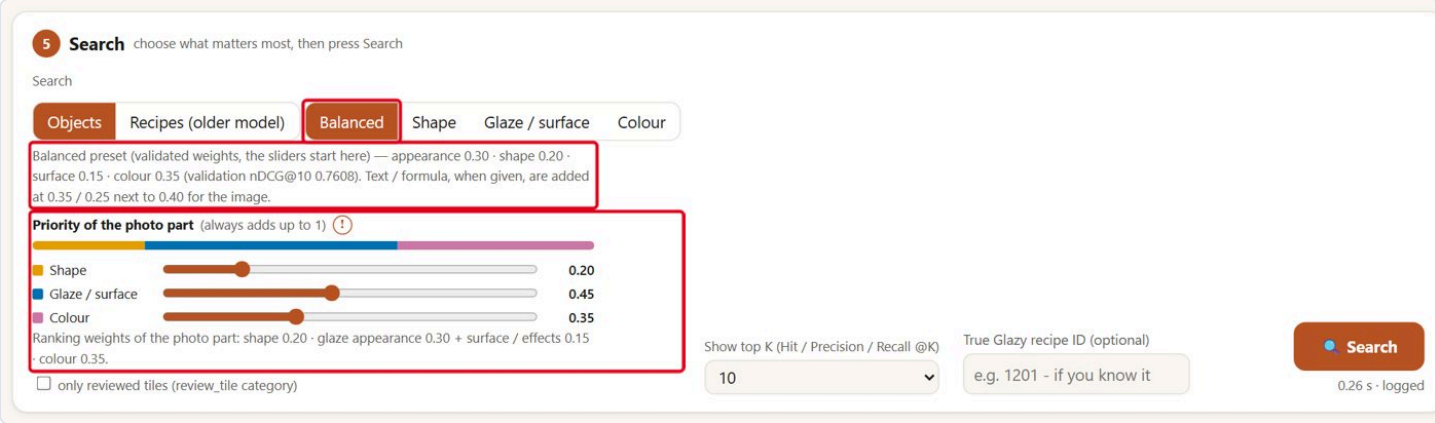
The evaluation table: the colour rule is within ΔE2000 20 of the typed colour.

Scenario 2 · The photo, Balanced preset (the default)

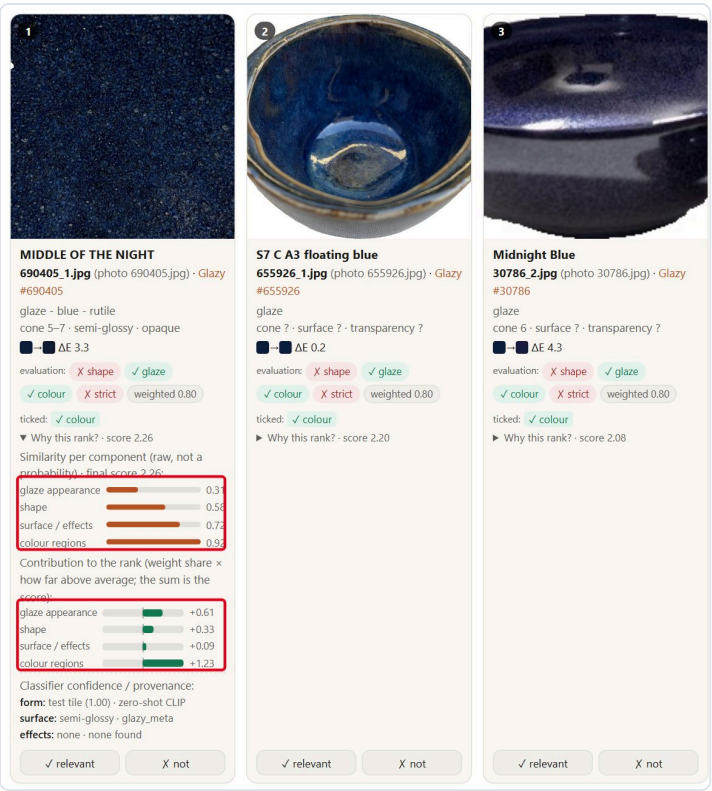
1. 📷 My photo: upload the photo. The app cuts out 1 object (the whole bowl).
2. Press 🏺 **Find similar pottery**: the page opens and searches at once with the Balanced preset.



The query: object #1 of your photo (framed).



Card 5 · Search: the Balanced preset and its weights, and the priority sliders of the photo part (framed).



Top 3; #1 opened: similarity per component (upper frame) and share × z (lower frame).

The calculation of #1, step by step

Shares: shape 0.20 · glaze appearance 0.30 · surface / effects 0.15 · colour regions 0.35.

component	similarity	average	spread	z	share	share × z
glaze appearance	0.313	-0.016	0.163	2.02	0.300	**+0.607**
shape	0.582	0.222	0.217	1.66	0.200	**+0.332**
surface / effects	0.718	0.642	0.129	0.59	0.150	**+0.088**
colour regions	0.919	0.227	0.197	3.52	0.350	**+1.231**
score of #1 MIDDLE OF THE NIGHT						**2.258**

Example, colour regions: z = (0.919 − 0.227) ÷ 0.197 = 3.52; × share 0.35 = +1.231. The four products add up to the score.

- **MIDDLE OF THE NIGHT** · 2.26
- **S7 C A3 floating blue** · 2.20
- **Midnight Blue** · 2.08

nDCG@10 **0.80** · Precision@10: shape 0 % · glaze 100 % · colour 100 % · strict 0 %

What it shows: #1 wins mainly on colour (similarity 0.92: nearly the same navy) and glaze look; all ten are glossy and in the colour.



The evaluation (framed): shape is checked against 'plate', the attribute model's guess for this bowl seen from above, so every result fails shape and strict is 0 %; colour uses the navy #0E1F3E; nDCG@10 weights the three by the priority sliders.

> **Read the evaluation with care.** It never changes the order; it checks each result with rules: shape = the same object form as your object, glaze = the same surface group, colour = within ΔE00 20 of your most important colour region, strict = all three. Here the form guess "plate" is wrong, so "shape ✗" and "strict 0 %" say more about the guess than about the results. Correct the form in card 1b (Your object up close) on 📷 My photo before searching to make those rows meaningful.

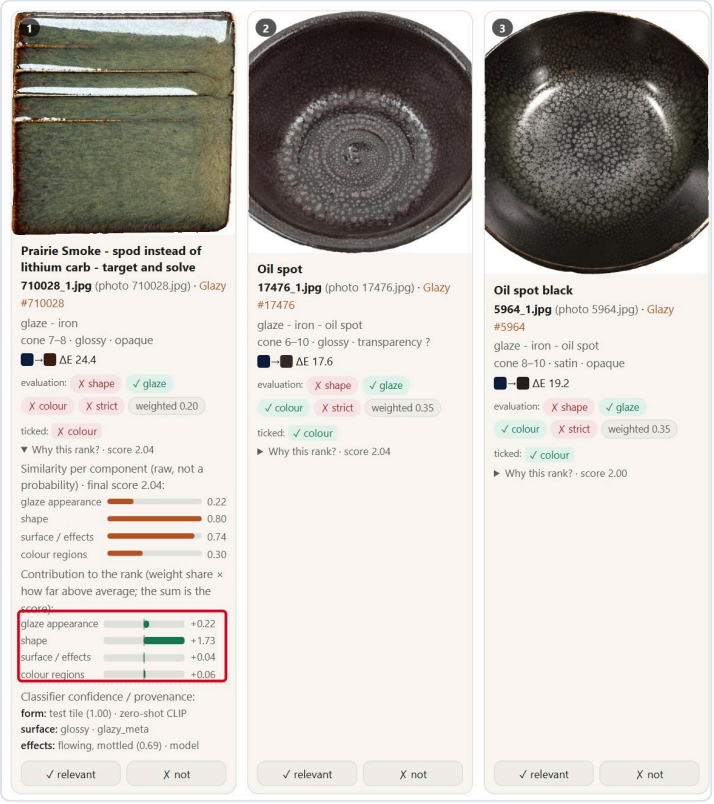
Scenario 3 · The same photo with the Shape, Glaze / surface and Colour presets

Click a preset in card 5: it searches again at once. Only the shares inside the photo part change.



The Shape preset (framed): shape now gets 0.65 of the photo part.

Preset	Shares (appearance · shape · surface · colour)	#1	Evaluation
Balanced	shape 0.20 · glaze appearance 0.30 · surface / effects 0.15 · colour regions 0.35	MIDDLE OF THE NIGHT	nDCG@10 **0.80** · Precision@10: shape 0 % · glaze 100 % · colour 100 % · strict 0 %
Shape	shape 0.65 · glaze appearance 0.15 · surface / effects 0.05 · colour regions 0.15	Prairie Smoke - spod instead of lithium carb - target and solve	nDCG@10 **0.38** · Precision@10: shape 10 % · glaze 100 % · colour 90 % · strict 10 %
Glaze / surface	shape 0.05 · glaze appearance 0.80 · surface / effects 0.05 · colour regions 0.10	GW10 base (magnesium matte)	nDCG@10 **0.45** · Precision@10: shape 10 % · glaze 50 % · colour 60 % · strict 10 %
Colour	shape 0.05 · glaze appearance 0.05 · surface / effects 0.10 · colour regions 0.80	MIDDLE OF THE NIGHT	nDCG@10 **0.94** · Precision@10: shape 0 % · glaze 90 % · colour 100 % · strict 0 %



Shape preset: the shape contribution dominates (framed). #1 is a test tile, #2 and #3 are bowls seen from above.

Shape preset, #1

component	similarity	average	spread	z	share	share × z
glaze appearance	0.220	-0.016	0.163	1.45	0.150	**+0.217**
shape	0.800	0.222	0.217	2.67	0.650	**+1.733**
surface / effects	0.739	0.642	0.129	0.74	0.050	**+0.037**
colour regions	0.303	0.228	0.196	0.38	0.150	**+0.057**
score of #1 Prairie Smoke - spod instead of lithium carb - target and solve						**2.045**

Shape now has the largest share, so outlines decide and colour hardly counts (#1 is green, ΔE 24 from the bowl). #1 is a **test tile**: the bowl fills almost the whole photo, so its outline is close to a full frame, like a tile photographed close up (shape similarity 0.80). #2 and #3 are bowls seen from above, the way this photo was taken. Shape compares the outline in the photo, not the real 3-D form.

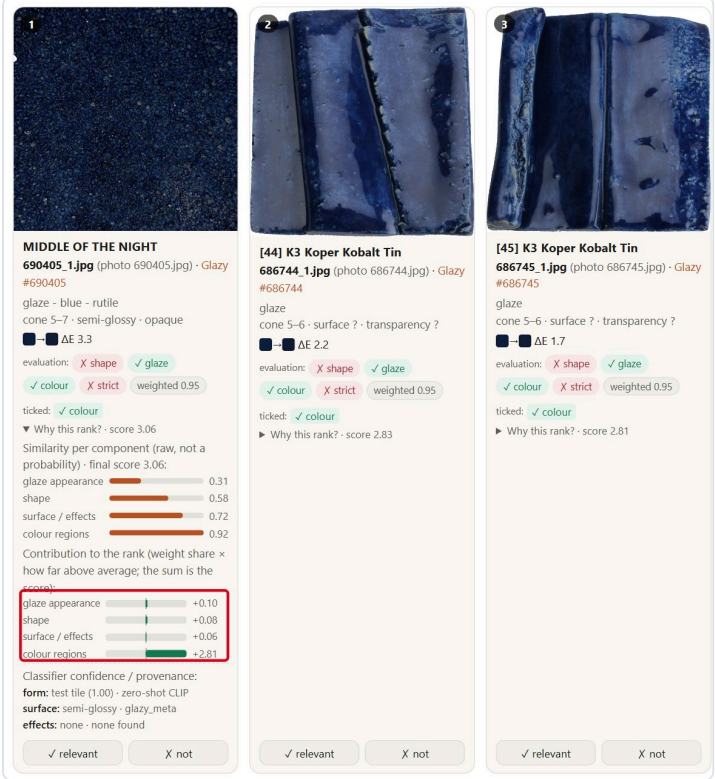


Glaze / surface preset: the glaze-look model decides (framed).

Glaze / surface preset, #1

component	similarity	average	spread	z	share	share × z
glaze appearance	0.586	-0.016	0.163	3.69	0.800	**+2.955**
shape	-0.102	0.222	0.217	-1.50	0.050	**-0.075**
surface / effects	0.441	0.642	0.129	-1.54	0.050	**-0.077**
colour regions	0.483	0.228	0.196	1.30	0.100	**+0.130**
score of #1 GW10 base (magnesium matte)						**2.932**

Glaze appearance gets 0.80 of the photo part. The look model puts two **magnesium matte** bases at #1 and #3 (GW10, GW11), with a floating cobalt at #2, although the bowl is glossy navy: this model compares texture and look more than colour, so with this preset check the colours.



Colour preset: the colour regions decide (framed).

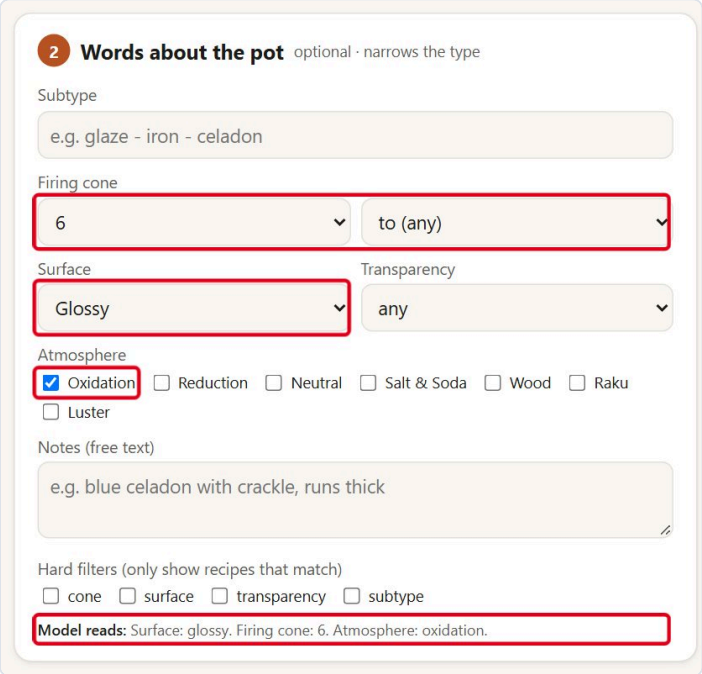
Colour preset, #1

component	similarity	average	spread	z	share	share × z
glaze appearance	0.313	-0.017	0.163	2.02	0.050	**+0.101**
shape	0.582	0.222	0.216	1.66	0.050	**+0.083**
surface / effects	0.718	0.642	0.129	0.59	0.100	**+0.059**
colour regions	0.919	0.228	0.196	3.52	0.800	**+2.813**
score of #1 MIDDLE OF THE NIGHT						**3.057**

Colour regions get 0.80 of the photo part. For this photo it gives the best evaluation (nDCG@10 0.94): the bowl is mostly one strong colour.

Scenario 4 · Photo + words

1. Card 2 · Words about the pot: firing cone **6**, surface **Glossy**, atmosphere **Oxidation**.
2. Press **Search** (Balanced).



Card 2: the words (framed) and what the model reads (bottom frame).

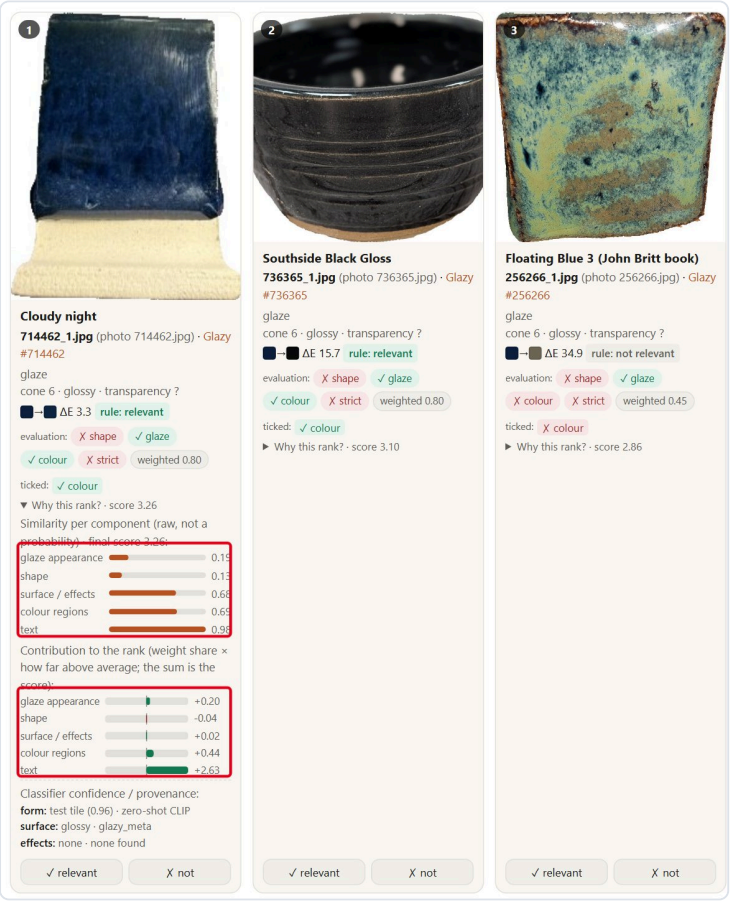
The words are cleaned into the training format by the text branch: `Surface: glossy. Firing cone: 6. Atmosphere: oxidation.`. The text component joins with share 0.35 next to 0.40 for the photo:

Shares: shape 0.11 · glaze appearance 0.16 · surface / effects 0.08 · colour regions 0.19 · text 0.47.

component	similarity	average	spread	z	share	share × z
glaze appearance	0.192	-0.017	0.163	1.28	0.160	**+0.205**
shape	0.133	0.222	0.217	-0.41	0.107	**-0.044**
surface / effects	0.679	0.642	0.129	0.29	0.080	**+0.023**
colour regions	0.689	0.227	0.197	2.35	0.187	**+0.438**
text	0.978	-0.007	0.174	5.65	0.467	**+2.635**
score of #1 Cloudy night						**3.258**

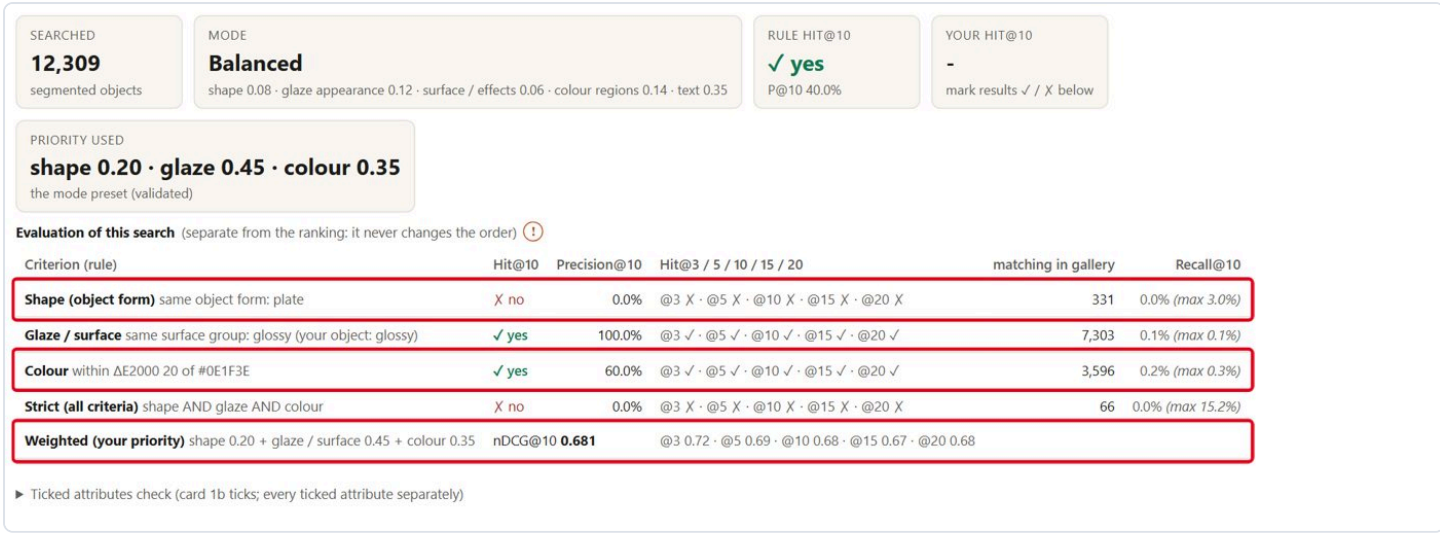
- **Cloudy night** · 3.26
- **Southside Black Gloss** · 3.10
- **Floating Blue 3 (John Britt book)** · 2.86

nDCG@10 **0.68** · Precision@10: shape 0 % · glaze 100 % · colour 60 % · strict 0 %



Top 3: the text contribution is the same for all three (framed): they share the same tags.

Why words make a photo search worse. Thousands of recipes carry exactly "glossy, cone 6, oxidation", so they all get the same very high text similarity (0.978, z ≈ 5.7). With 47 % of the score, the words then decide the order and push the photo's evidence aside: nDCG@10 falls from 0.80 (scenario 2) to 0.68, and colour precision from 100 % to 60 %. **With a photo, leave the words empty**; use them when you have no photo.



The evaluation after adding the words (framed rows).

Scenario 5 · Photo + words + formula

1. Keep the words of scenario 4.
2. Card 3 · Formula: paste the formula of a navy glossy glaze (Storm Blue, Glazy #577829, from the recipe catalog) and press **Fill**: 11 oxides including fluxes, so it is used.
3. Press **Search**.



Card 3: the oxides (framed) and the check that the formula is used (lower frame).

Shares: shape 0.08 · glaze appearance 0.12 · surface / effects 0.06 · colour regions 0.14 · text 0.35 · formula 0.25 (photo 0.40 + words 0.35 + formula 0.25 = 1).

component	similarity	average	spread	z	share	share × z
glaze appearance	0.326	-0.017	0.163	2.10	0.120	**+0.252**
shape	0.344	0.222	0.217	0.56	0.080	**+0.045**
surface / effects	0.781	0.642	0.129	1.07	0.060	**+0.064**
colour regions	0.535	0.228	0.197	1.56	0.140	**+0.219**
text	0.721	-0.006	0.174	4.17	0.350	**+1.461**
formula	0.330	-0.001	0.207	1.60	0.250	**+0.400**
score of #1 Frogpond blue						**2.442**

- **Frogpond blue** · 2.44
- **Cloudy night** · 2.36
- **Floating Blue 3 (John Britt book)** · 2.18

nDCG@10 **0.68** · Precision@10: shape 0 % · glaze 100 % · colour 60 % · strict 0 %

What it shows: the formula lifts recipes whose chemistry is close (Frogpond blue: formula similarity 0.33, the highest of the three) and takes some weight from the words. The words still dominate, so with a photo the best results stay in scenarios 2–3.



Frogpond blue

70574_1.jpg (photo 70574.jpg) · Glazy #70574

glaze

cone 5 1/2–7 · glossy · transparency ?

→ ΔE 9.1 **rule: relevant**

evaluation: ✗ shape ✓ glaze ✓ colour ✗ strict weighted 0.80

ticked: ✓ colour

▼ Why this rank? · score 2.44

Similarity per component (raw, not a probability) · final score 2.44:

glaze appearance	0.33
shape	0.34
surface / effects	0.78
colour regions	0.54
text	0.72
formula	0.33

Contribution to the rank (weight share × how far above average; the sum is the score):

glaze appearance	+0.25
shape	+0.04
surface / effects	+0.06
colour regions	+0.22
text	+1.46
formula	+0.40

Classifier confidence / provenance:

form: test tile (1.00) · zero-shot CLIP

surface: glossy · glazy_meta

effects: flowing (0.47) · model

✓ relevant ✗ not



Cloudy night

714462_1.jpg (photo 714462.jpg) · Glazy #714462

glaze

cone 6 · glossy · transparency ?

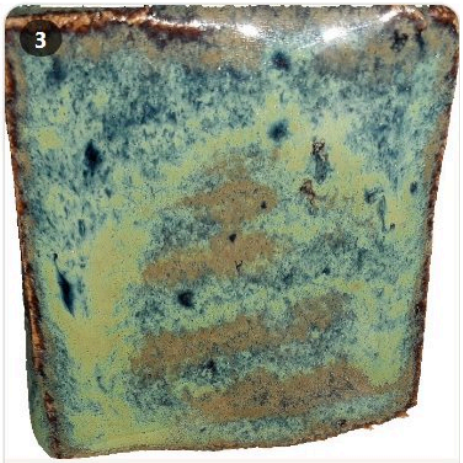
→ ΔE 3.3 **rule: relevant**

evaluation: ✗ shape ✓ glaze ✓ colour ✗ strict weighted 0.80

ticked: ✓ colour

▶ Why this rank? · score 2.36

✓ relevant ✗ not



Floating Blue 3 (John Britt book)

256266_1.jpg (photo 256266.jpg) · Glazy #256266

glaze

cone 6 · glossy · transparency ?

→ ΔE 34.9 **rule: not relevant**

evaluation: ✗ shape ✓ glaze ✗ colour ✗ strict weighted 0.45

ticked: ✗ colour

▶ Why this rank? · score 2.18

✓ relevant ✗ not

Top 3: #1 opened: text, formula and the four photo components (framed).



The evaluation with words and formula (framed rows).

Summary

Scenario	Inputs	Shares	#1	nDCG@10	glaze P@10	colour P@10
1	a colour only	colour regions 1.00	MIDDLE OF THE NIGHT (3.52)	1.00	–	100 %
2	photo · Balanced	shape 0.20 · glaze appearance 0.30 · surface / effects 0.15 · colour regions 0.35	MIDDLE OF THE NIGHT (2.26)	0.80	100 %	100 %
3a	photo · Shape	shape 0.65 · glaze appearance 0.15 · surface / effects 0.05 · colour regions 0.15	Prairie Smoke - spod instead of lithium carb - target and solve (2.04)	0.38	100 %	90 %
3b	photo · Glaze / surface	shape 0.05 · glaze appearance 0.80 · surface / effects 0.05 · colour regions 0.10	GW10 base (magnesium matte) (2.93)	0.45	50 %	60 %
3c	photo · Colour	shape 0.05 · glaze appearance 0.05 · surface / effects 0.10 · colour regions 0.80	MIDDLE OF THE NIGHT (3.06)	0.94	90 %	100 %
4	photo + words	shape 0.11 · glaze appearance 0.16 · surface / effects 0.08 · colour regions 0.19 · text 0.47	Cloudy night (3.26)	0.68	100 %	60 %
5	photo + words + formula	shape 0.08 · glaze appearance 0.12 · surface / effects 0.06 · colour regions 0.14 · text 0.35 · formula 0.25	Frogpond blue (2.44)	0.68	100 %	60 %

- **A photo alone** works best here; the **Colour** preset gives the best evaluation for a pot that is mostly one strong colour, **Shape** finds the same form, **Glaze / surface** finds the same glaze look (check the colours).
- **Words with a photo** make the order worse, because thousands of recipes share the same tags; a **formula** softens that a little.
- The score is a sum of shares × z: 0 is an average pot. The card's *Why this rank?* shows each part.
- The evaluation is only as good as the object's form and surface: correct them in card 1b before trusting "shape" and "strict".